Notes on Topology and Manifolds

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Based on Munkres' Topology *and Tu's* Introduction to Manifolds

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Preface

These notes were compiled during my studies of topology and manifolds, primarily drawing from James Munkres' *Topology* and Loring Tu's *Introduction to Manifolds*. Initially intended as a personal resource to solidify my understanding, they have since evolved into a more comprehensive document that I am sharing publicly to benefit others pursuing similar topics.

Please note that these notes are an ongoing project and will be regularly updated and refined as my studies progress. My aim is to ensure accuracy, clarity, and depth in the presentation of the material. However, as these notes remain a work in progress, they may contain gaps or areas for improvement.

I welcome suggestions, corrections, or any feedback that could improve the quality of this document. Please feel free to reach out to me via Discord at `@j.en` or by email at `vorahsan@gmail.com`. Your input is highly appreciated.

– *Rafid Ahsan*

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Chapter 1

Set Theory

1.1 Introduction

In mathematics, we encounter a wide variety of objects. Set theory provides a fundamental framework for constructing these objects, while logic helps us distinguish and describe any equivalent relationships between them. In this section, we will acquaint ourselves with the essential concepts of set theory and the basics of logical reasoning.

1.2 Sets

Definition 1.2.1: Set

A **set** is a collection of objects, called **elements**. We denote a set by curly braces, $\{\}$, and list the elements inside.

- If an item a belongs to the set P, we denote it by $a \in P$. If an item b does not belong to the set P, we denote it by $b \notin P$.
- A set may be constructed using **set builder notation** as follows. Suppose P is our set. Then, we can write

$$P = \{x \mid \text{some condition } C \text{ is true for } x\},$$

  where x is an element from a specified domain. This is read as "The set of all x such that some condition C is satisfied."
- We denote the **empty set** by $\emptyset$ or $\{\}$.

Example 1.2.1:
Some common examples of sets are:

- The set of natural numbers: $\mathbb{N} = \{1, 2, 3, \ldots\}$.
- The set of integers: $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
- The set of rational numbers: $\mathbb{Q} = \left\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\right\}$.
- The set of real numbers: $\mathbb{R} = \{x \mid x \text{ is a real number}\}$.

△

Example 1.2.2:
The set $\mathcal{V}$ of all English vowels can be written as

$$\mathcal{V} = \{a, e, i, o, u\}.$$

△

Example 1.2.3:
A ***collection*** refers to a set whose *elements are themselves sets*. Consider the following sets:

$$E = \{2, 4, 6, 8, \dots\}, \quad O = \{1, 3, 5, 7, \dots\}, \quad P = \{2, 3, 5, 7, 11, \dots\}$$

where:

- E is the set of all **even numbers**,
- O is the set of all **odd numbers**, and
- P is the set of all **prime numbers**.

We can now define a new set X as a *collection of sets*:

$$X = \{E, O, P\}$$

Here, each element of X is itself a set. Thus, X is a *collection of sets*. For example, the element $E \in X$ represents the set of all even numbers.

Notice that:

$$x \in X \implies x \text{ is a set.}$$

Sets are not limited to containing only numbers. The preceding examples highlight that sets can consist of various types of objects, including numbers, letters, or even other sets!

△

Definition 1.2.2: Subset

A set A is a **subset** of a set B if every element of A is also an element of B. We denote this by $A \subseteq B$. This is read as "A is a subset of B" or "A is contained in B," which is the convention we will use throughout these notes.

Example 1.2.4:
Here are some common examples of subsets:

- The set of even numbers is a subset of natural numbers: $E \subseteq \mathbb{N}$, where $E = \{2, 4, 6, 8, \dots\}$.
- The set of integers is a subset of real numbers: $\mathbb{Z} \subseteq \mathbb{R}$.
- The set of rational numbers is a subset of real numbers: $\mathbb{Q} \subseteq \mathbb{R}$.
- The empty set is a subset of every set: $\emptyset \subseteq A$ for any set A.

△

Example 1.2.5:
Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$. Then A is a subset of B, denoted as $A \subseteq B$, since every element of A is also in B. Similarly, B is not a subset of A, denoted as $B \not\subseteq A$, since 4 and 5 are elements of B but not in A.

△

Theorem 1.2.1: Set Equality

Two sets A and B are equal if and only if they have the same elements. We denote this by $A = B$.

Proof: Let A and B be two sets. Assume $A \subseteq B$ and $B \subseteq A$. Therefore, every element in A is also in B, and vice versa. Hence, $A = B$. □